Metal Prices in Transition: Copper, Silver, and Structural Tightness Progress Presentation

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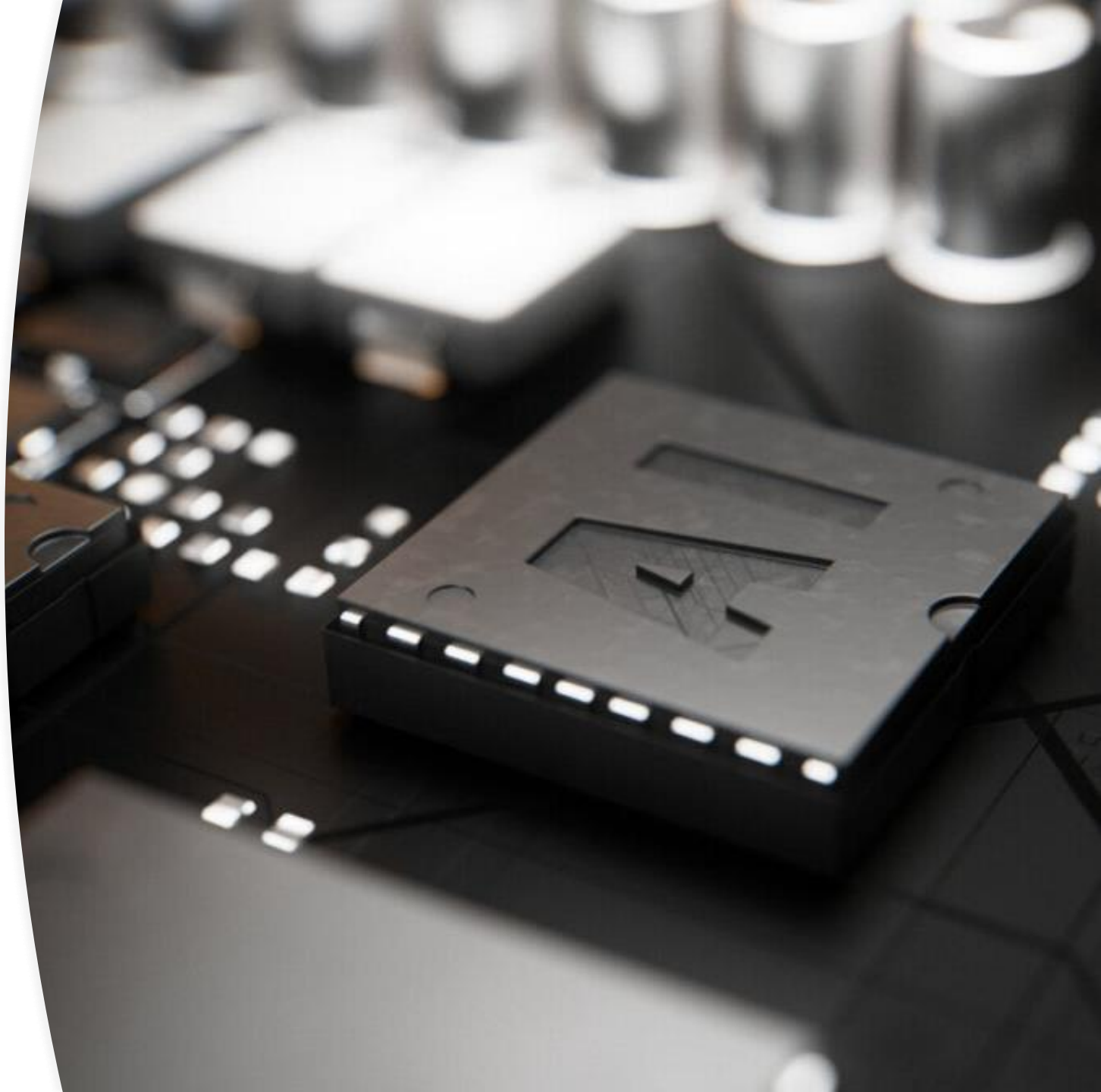
Why this topic matters

- Metal markets sit at the intersection of macroeconomics, industrial production, and the energy transition
- Copper matters for electrification, grids, EVs, and infrastructure
- Silver matters because it is both a precious metal and an industrial input, especially in solar and electronics
- That makes these two metals especially interesting to study together

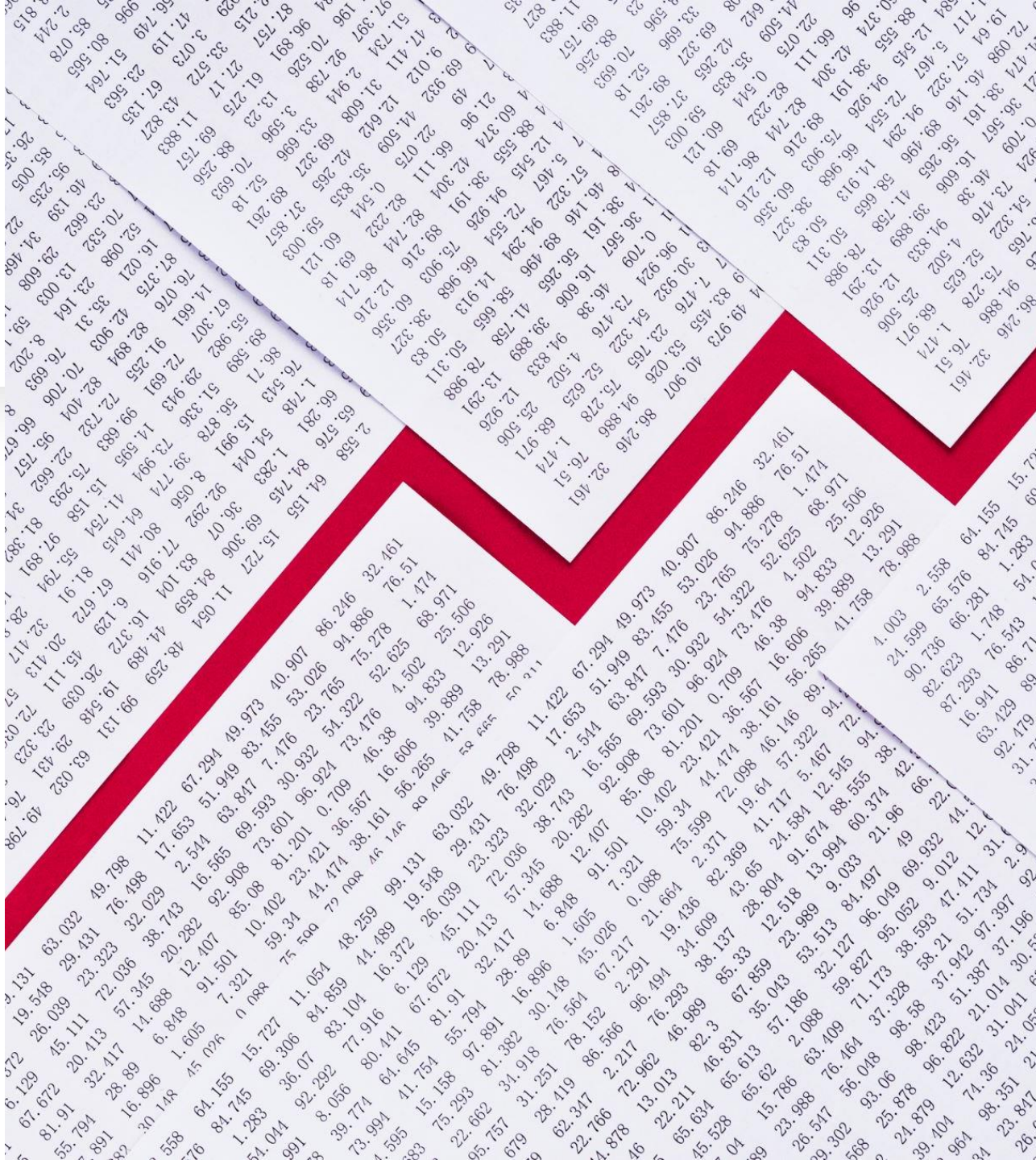
Research question

- **Main question:** Are copper and silver becoming structurally tighter than other major metals?
- **Secondary question:** How much of metal price movement can be explained by macro conditions versus physical-market tightening?
- **Goal:** combine historical price evidence with structural market evidence in a way that is transparent and academically defensible



Data and scope

- Monthly metal prices from the World Bank Pink Sheet
- Macro controls from FRED, including VIX, yield spread, unemployment, S&P 500, and the broad dollar
- Additional physical-market inputs from LME/Westmetall, IEA, ICSG, Silver Institute, OECD, and USGS
- Main comparison set: aluminum, copper, nickel, zinc, gold, and silver
- Historical sample spans roughly 2000–2026, with regression work centered on the most consistent monthly window



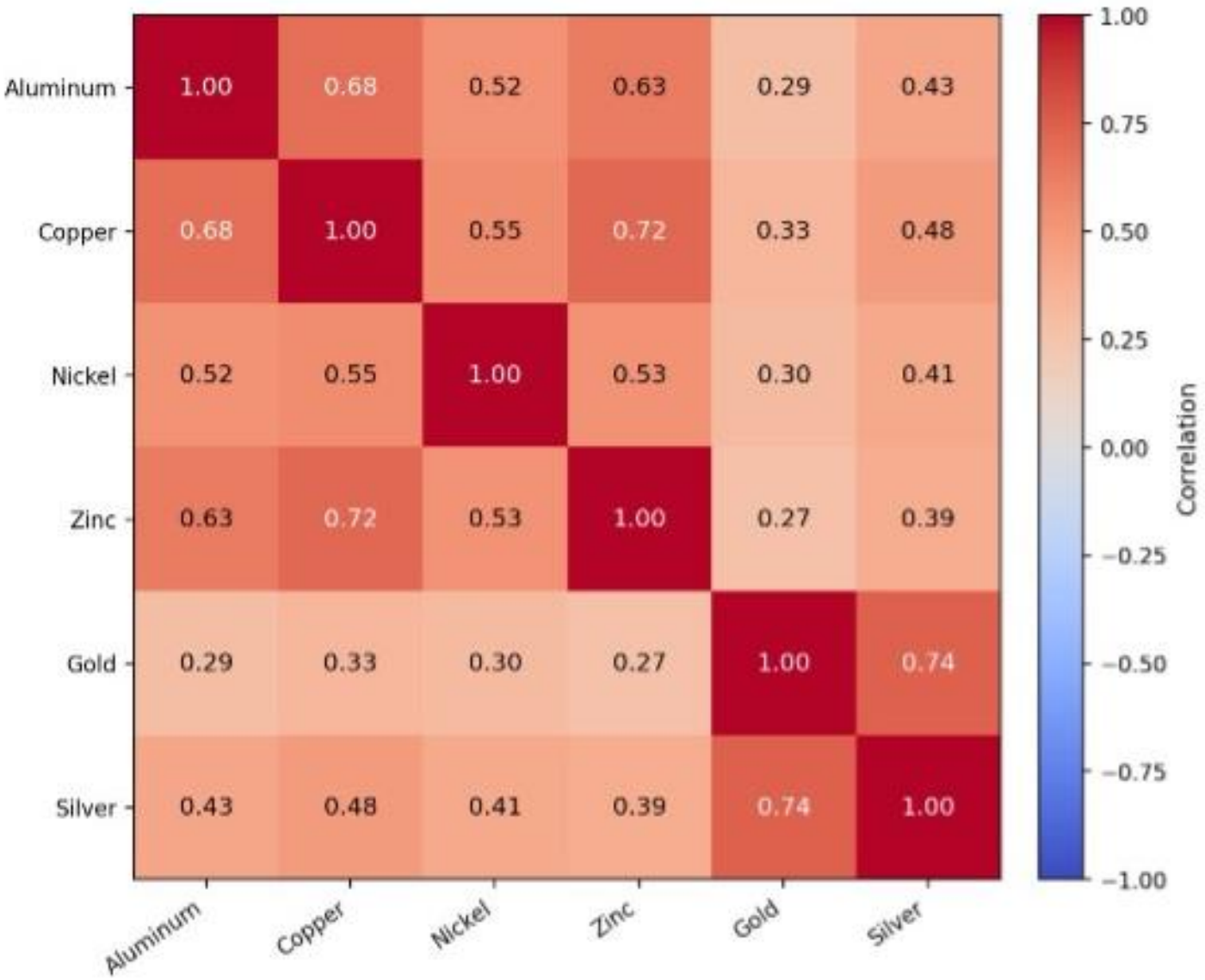
Aluminum
Copper
Nickel
Zinc
Gold
Silver
Index
1,750
1,500
1,250
1,000
750
500
250
0
2000
2004
2008
2012
2016
2020
2024

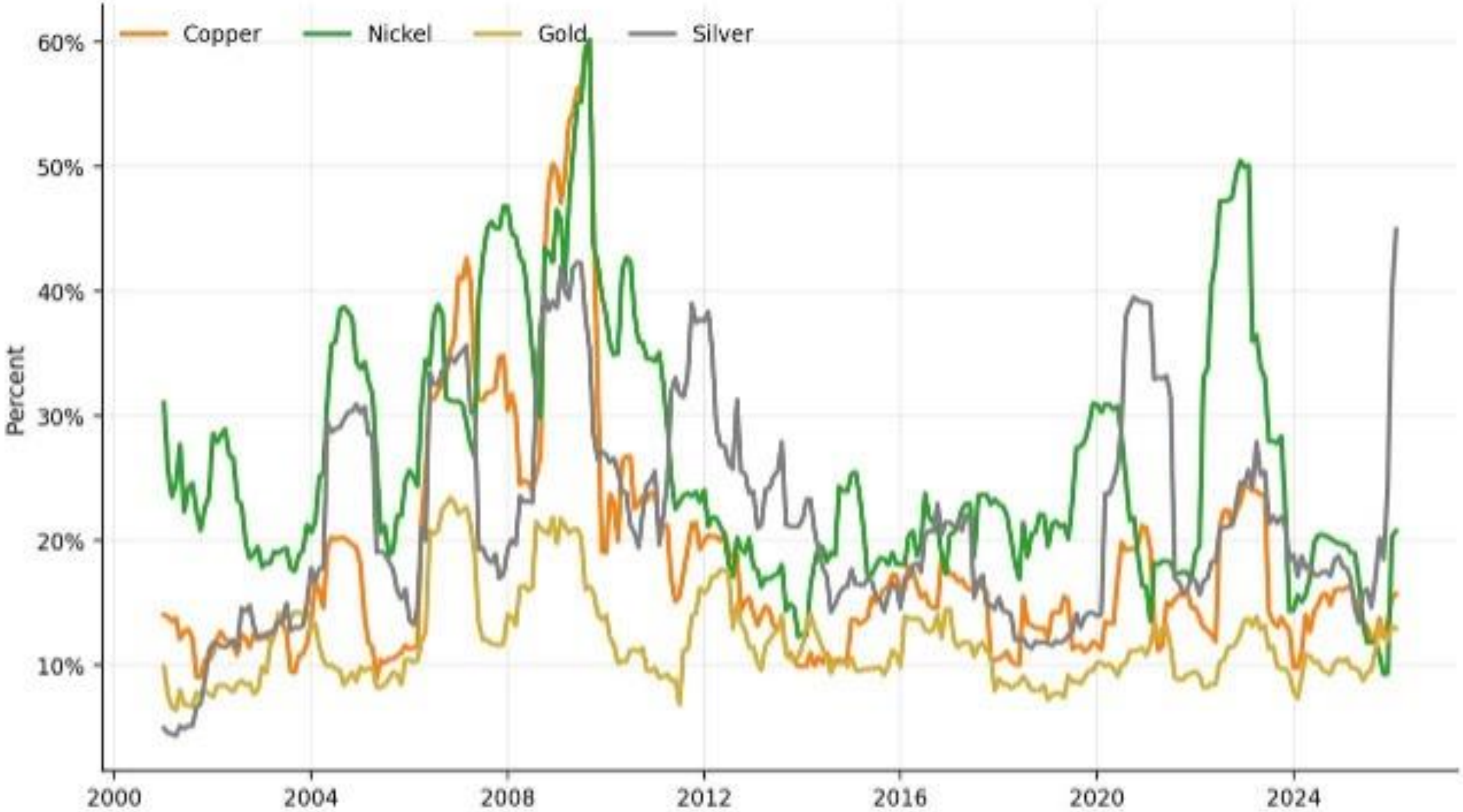
COMEX SILVER INVENTORY: MARCH 12, 2026 VS. ISSUE #9 (FEB 27, 2026)

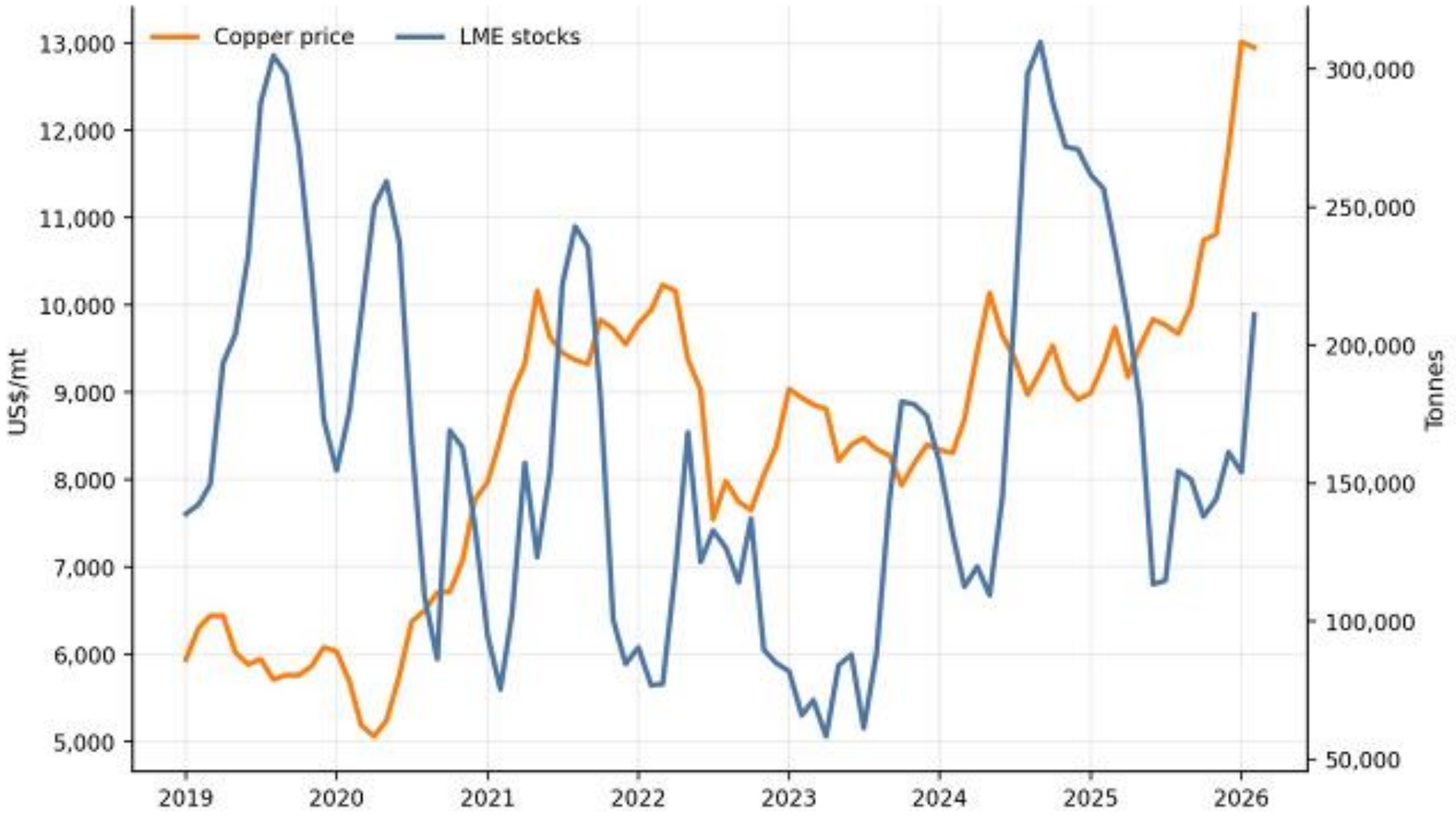
CATEGORY	MARCH 12, 2026	FEB 27, 2026 (ISSUE #9)	CHANGE
Registered	**78.95 Moz**	88.78 Moz	**–9.83 Moz**
Eligible	262.77 Moz	~278 Moz	**–15.2 Moz**
Combined Total	**341.72 Moz**	**~366 Moz**	**–24.3 Moz**

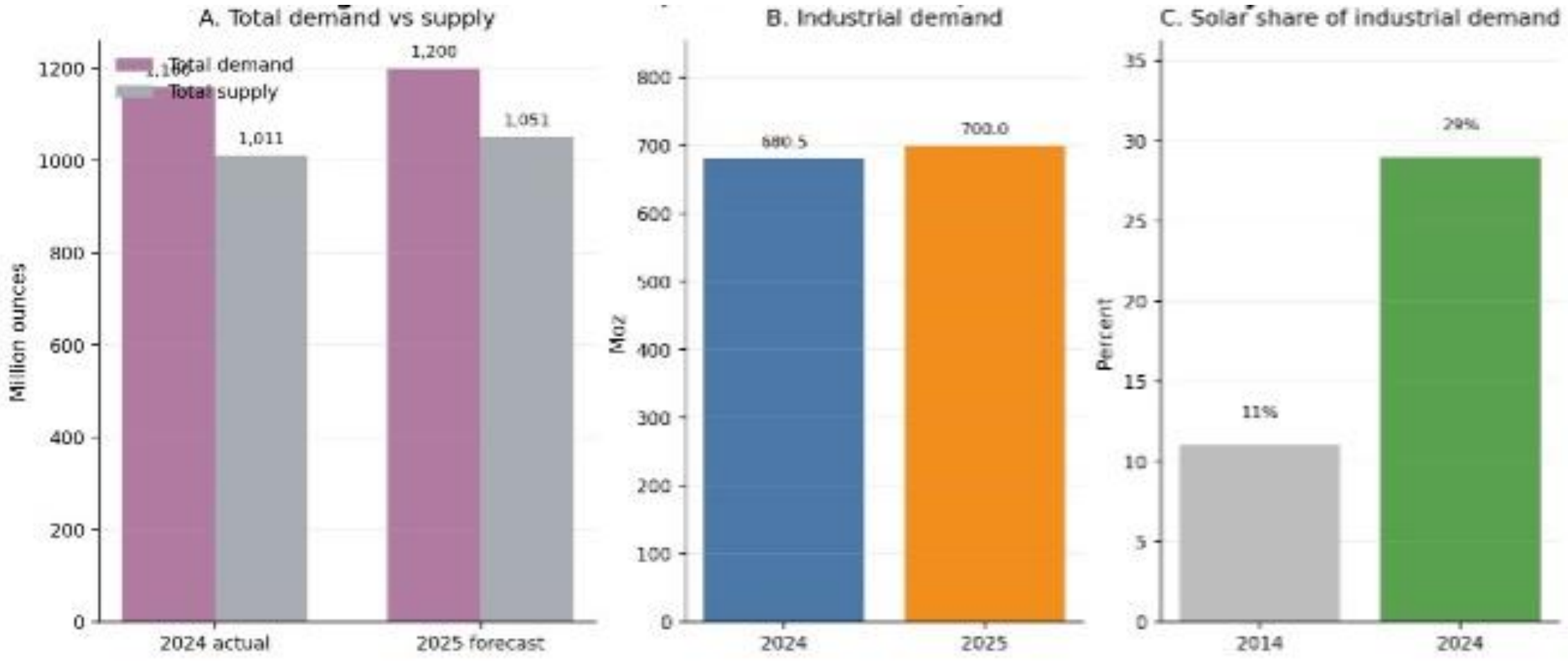
Descriptive statistics, January 2000–February 2026.

Metal	Start price	End price	CAGR %	Mean monthly return %	Std. monthly return %	Annualized vol %
Aluminum	$1680.28	$3065.19	2.33 %	0.19 %	4.9 %	16.99 %
Copper	$1843.98	$12951.35	7.76 %	0.62 %	6.15 %	21.29 %
Nickel	$8309.5	$17173.1	2.82 %	0.23 %	8.41 %	29.15 %
Zinc	$1178.8	$3323.71	4.05 %	0.33 %	6.45 %	22.35 %
Gold	$284.32	$5019.97	11.64 %	0.92 %	3.67 %	12.7 %
Silver	$5.19	$81.95	11.16 %	0.88 %	7.11 %	24.62 %


Aluminum
Copper
Nickel
Zinc
Gold
Silver
1.00 0.68 0.52 0.63 0.29 0.43
0.68 1.00 0.55 0.72 0.33 0.48
0.52 0.55 1.00 0.53 0.30 0.41
0.63 0.72 0.53 1.00 0.27 0.39
0.29 0.33 0.30 0.27 1.00 0.74
0.43 0.48 0.41 0.39 0.74 1.00
Aluminum
Copper
Nickel
Zinc
Gold
Silver
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Correlation


Copper
Nickel
Gold
Silver
60%
50%
40%
30%
20%
10%
Percent
2000
2004
2008
2012
2016
2020
2024


Copper price
LME stocks
US$/mt
Tonnes
13,000
12,000
11,000
10,000
9,000
8,000
7,000
6,000
5,000
300,000
250,000
200,000
150,000
100,000
50,000
2019
2020
2021
2022
2023
2024
2025
2026


A. Total demand vs supply
Total demand
Total supply
Million ounces
1,166
1,011
1,200
1,051
2024 actual
2025 forecast
B. Industrial demand
Moz
680.5
700.0
2024
2025
C. Solar share of industrial demand
Percent
11%
29%
2014
2024

Current progress

- Collected and cleaned the main monthly price data
- Merged the macro controls into the historical panel
- Built copper- and silver-specific structural input tables
- Created a clear paper structure and figure sequence
- Identified the main limitation: physical-balance data are more fragmented than price data, so assumptions need to be stated carefully

Ethical, Legal, and Societal Implications

1. Ethical Sourcing and Labor Practices

- Rising demand for industrial metals increases pressure on mining regions, many of which face labor exploitation, unsafe working conditions, and weak environmental protections.
- Stakeholders face growing ethical responsibility to ensure transparent and responsible sourcing standards.

2. Environmental and Sustainability Concerns

- Expanded mining activity required to meet energy-transition demand can lead to land degradation, water pollution, and biodiversity loss.
- Balancing climate goals with environmental protection creates a major ethical trade-off for governments and industry.

3. Resource Inequality and Community Impact

- Mining benefits are often unevenly distributed, while local communities bear environmental and social costs.
- Societal tensions may increase if economic gains do not translate into local development, infrastructure, or fair compensation.

4. Regulatory and Legal Compliance Pressure

- Governments are increasingly introducing regulations on emissions, waste management, and supply-chain transparency.
- Companies face growing legal risks related to ESG disclosure requirements, environmental liability, and human-rights compliance.

5. Global Economic and Geopolitical Stability

- Concentration of critical mineral supply in a few countries raises risks of trade disputes, export controls, and geopolitical leverage.
- Societies dependent on energy transition technologies may face price shocks or supply disruptions affecting economic stability.

Key Implications for Stakeholders

1. Investors — Portfolio Diversification Opportunities

- Industrial metals move closely together, while gold and silver follow partially different drivers.
- Investors can improve risk diversification by combining precious metals with base metals rather than concentrating within industrial commodities alone.
- Silver offers a hybrid exposure to both economic growth and monetary hedging.

2. Mining Companies — Strategic Production Planning

- Strong correlations among base metals imply shared demand cycles tied to global industrial activity.
- Producers face synchronized price risks during economic downturns.
- Firms may benefit from diversifying production portfolios across both industrial and precious metals to stabilize revenues.

3. Manufacturers & Industrial Users — Input Cost Volatility Risk

- High co-movement among industrial metals means cost pressures can rise simultaneously across multiple inputs.
- Companies relying on copper, aluminum, nickel, or zinc should consider hedging strategies or long-term supply contracts.
- Energy transition demand may amplify future price volatility.

4. Policymakers — Resource Security and Energy Transition Risks

- Rising structural demand for metals linked to electrification and renewable technologies increases supply vulnerability.
- Persistent deficits or tight inventories could create strategic resource dependencies.
- Governments may need to support recycling, domestic mining, or critical mineral supply chains.

5. Financial Markets & Risk Managers — Changing Commodity Dynamics

- Precious metals retain safe-haven characteristics, while industrial metals remain growth-sensitive.
- Correlation structures imply commodities respond differently across inflation, recession, and monetary policy regimes.
- Risk models must account for metals behaving as **distinct macro asset classes**, not a single commodity group.

Next Steps

- FINALIZE THE MODEL SPECIFICATION
- REFINE THE STRUCTURAL SECTION SO IT STAYS CAREFUL AND CREDIBLE
- IMPROVE THE VISUAL FLOW BETWEEN HISTORICAL EVIDENCE AND PHYSICAL-MARKET EVIDENCE
- PREPARE THE FINAL POSTER, AND FINAL PRESENTATION